

Physics Test Paper - Mechanics

Subject: Physics | Chapter: Mechanics | Difficulty: medium

Question 1: A car of mass 1000 kg accelerates from rest to 20 m/s in 10 seconds. Ignoring friction, what is the average power delivered by the car's engine? (Use the formula: $\text{Power} = \text{Work}/\text{Time}$, and $\text{Work} = 0.5 \cdot m \cdot v^2$)

Options:

- A) 20 kW
- B) 200 kW
- C) 2000 kW
- D) 20000 kW

Question 2: A ball is thrown vertically upwards with an initial velocity of 20 m/s. Ignoring air resistance, approximately how high will it reach before falling back down? (Use the formula: $v^2 = u^2 + 2as$, where v = final velocity, u = initial velocity, a = acceleration due to gravity (-9.8 m/s^2), and s = displacement)

Options:

- A) 10 m
- B) 20 m
- C) 40 m
- D) 100 m

Question 3: A crane lifts a 500 kg load vertically upwards at a constant speed of 1 m/s. What is the power required by the crane's motor, assuming 100% efficiency? (Use the formula: $\text{Power} = \text{Force} \times \text{Velocity}$, and $\text{Force} = \text{mass} \times \text{acceleration due to gravity}$ ($g = 9.8 \text{ m/s}^2$))

Options:

- A) 490 W
- B) 4900 W
- C) 49000 W
- D) 490000 W

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